Tech-Edge Lab

Topic: -

Generative AI & Intelligent Application Development Bootcamp

Sub-topic:-

Industry-Oriented Training Program on LLMs, AI Tools, AI Agents & Vibe Coding

Module 1:- AI industry and Careers

Lecture 2:- AI Ecosystem: Understanding AI, ML, Deep Learning, NLP, Computer Vision, Robotics & Generative AI

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Recap: Introduction to AI World

- **Yesterday We Learned:**
  - Why AI is important
  - AI career opportunities
  - AI vs Traditional Programming
  - Role of AI in software development

- **Today:**
  - How different AI technologies work together.

Activity

Name 3 AI applications you used today.

Name	AI Feature used in that

What is an Ecosystem?

- **Definition**

  An ecosystem is a group of technologies that work together to create intelligent systems.

- **Human Intelligence Example:**
  - Human uses:
  - Eyes → Seeing
  - Ears → Listening
  - Brain → Thinking
  - Memory → Learning
  - Decision Making → Action

- AI systems also combine multiple technologies.

```
Artificial Intelligence
                   |
      --------------------------------
      |                       |
Machine Learning          Rule Based AI
      |
Deep Learning
      |
-------------------------
|                 |
NLP             Computer Vision
|
Generative AI
|
AI Applications
|
Robotics
```

Important Points

AI is the biggest field.
ML is a technique to achieve AI.
Deep Learning is a powerful ML method.
NLP helps machines understand language.
Computer Vision helps machines see.
Generative AI creates new content.
Robotics combines AI with machines.

- Human:
  - **Parts → Field of AI**
  - Eyes →?
  - Brain → ?
  - Memory → ?
  - Learning → ?
  - Language → ?
  - Decision → ?

Answers

- Eyes → Computer Vision
- Brain → AI
- Memory → Data
- Learning → Machine Learning
- Language → NLP
- Decision → AI Algorithms

Activity 1

- **Identify the Technology**

  students:

1) Face Unlock

- Answer:

  ?

2) Google Translate

- Answer:

  ?

3) ChatGPT

- Answer:

  ?

4) Self-driving Car

- Answer:

  ?

Answers

- **Face Unlock**

  Answer:
  Computer Vision + Deep Learning
- **Google Translate**

  Answer:
  NLP
- **ChatGPT**

  Answer:

  NLP + Deep Learning + Generative AI
- **Self-driving Car**

  Answer:

  Computer Vision + ML + Robotics

Artificial Intelligence (AI)

- **Definition**

  Artificial Intelligence enables machines to perform tasks that normally require human intelligence.

- **AI Capabilities**
  - Learning
  - Reasoning
  - Planning
  - Decision Making
  - Problem Solving

- **Examples**
  - Google Maps
  - Siri
  - Alexa
  - ChatGPT
  - Recommendation Systems

- **Activity 2**

  Is it AI or Not?

- Example AI? (Yes/No)

  Calculator

  Google Maps

  YouTube Recommendation

  Simple ATM Machine

  ChatGPT

  Snapchat Filters

  TV Remote control

Example	Is It AI ?
Calculator	No
Google Aps	Yes
You tube recommendation	Yes
Simple ATM machine	No
Chat Gpt	Yes
Snapchat Filters	Yes
TV Remote Control	No

Types of AI

1. **Artificial Narrow Intelligence (ANI):-** Artificial Narrow Intelligence (ANI), or weak AI, refers to AI systems designed to excel at a single, highly specific task.

Examples:

Most AI today.

ChatGPT

Google Translate

Alexa

2. **Artificial General Intelligence (AGI)**

AI system which develop Human-level intelligence.

Not achieved yet.

3. **Artificial Super Intelligence (ASI)**

AI beyond human intelligence.

Future concept.

Machine Learning

Definition

Machine Learning allows computers to learn from data without being explicitly programmed.

Traditional Programming

Rules + Data (C tokens, function, keywords)

Program

Output (sum.c , cal.c)

- **Machine Learning**

  Data + Output Examples (C tokens, function, keywords + examples)

  ML Model (supervised, unsupervised, reinforcement)

  Learn Rules

Example: Spam Detection

Traditional Approach:

Create rules:

- Contains "free money"
- Unknown sender
- Suspicious links

Problem:

* Too many rules.

Machine Learning:-

- Give thousands of emails.
- AI learns patterns.

Types of Machine Learning

1. Supervised learning
2. Unsupervised learning
3. Reinforcement learning’

Supervised Learning

Learning with labelled data.

Examples:

- Spam Detection
- Disease Prediction
- Student Result Prediction

Supervised learning is exactly like studying for an exam using a practice quiz that includes the answer key.

Unsupervised Learning

Finding hidden patterns.

Examples:

Customer Segmentation

Shopping Behavior

- Unsupervised learning is like walking into a massive library where none of the books have covers, titles, or genres written on them—and there is no librarian or answer key to help you.

- Instead of checking an answer key, the AI must explore the books on its own, find similarities in how they are written, and automatically group them together (e.g., grouping books with magic spells into one pile, and books with spaceships into another).

Reinforcement Learning

Learning through rewards.

Examples:

- Game AI
- Robots
- Self-driving cars

- Reinforcement learning is exactly like training a puppy to do a trick using treats and time-outs.

- The AI does not have an answer key, and it does not have a teacher telling it what to do. Instead, it is dropped into an environment and forced to learn through trial and error. When it does something right, it gets a "reward" (a digital treat). When it crashes or fails, it gets a "punishment" (a penalty score).

Activity 3

Q- Which ML type is used?

1) Predict house price

Answer:

?

2) Group customers based on shopping

Answer:

?

3) Robot learning walking

Answer:

?

Answers:-

Predict house price

Answer:

Supervised

Group customers based on shopping

Answer:

Unsupervised

Robot learning walking

Answer:

Reinforcement

Machine Learning Workflow

Collect Data

↓

Clean Data

↓

Train Model

↓

Test Model

↓

Deploy

↓

Prediction

Example:

Student Placement Prediction

Input:

CGPA

Skills

Attendance

Projects

Output:

Placed / Not Placed

Deep Learning

Definition

Deep Learning is a subset of Machine Learning that uses artificial neural networks.

Inspired by human brain.

Used for:

- Images
- Voice
- Video
- Language

Examples:

- Face Recognition
- ChatGPT
- Image Generation

Artificial Neural Network

Human Brain:

Neuron → Connection → Learning

AI:

Input Layer

↓

Hidden Layers

↓

Output Layer

More layers = Deep Learning

Human Brain vs. Artificial Neural Network

Different in form, Similar in Function

Neurons
Basic processing units of the brain

Synapses
Connections between neurons to transmit signals

Learning
From experience and feedback

Memory
Stores knowledge and experiences

Adaptation
Adjusts and evolves over time

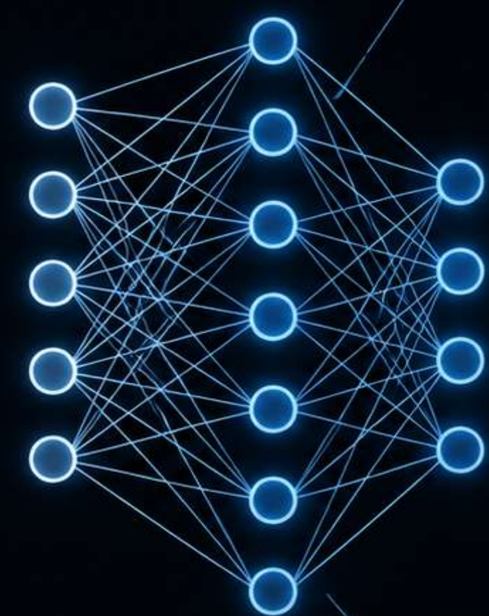
SIMILARITIES

Information Processing
Both process information

Learning
Both learn from data/experience

Memory
Both store knowledge

Adaptation
Both improve over time



Nodes
Basic processing units (like neurons)

Weights
Connections between nodes (like synapses)

Learning
From training data

Memory
Stored in weights and biases

Adaptation
Improves with more data

HUMAN BRAIN	ARTIFICIAL NEURAL NETWORK
~86 billion neurons	Millions to billions of artificial neurons
Synapses are chemical	Connections are numerical (weights)
Highly parallel	Parallel but limited by hardware
Works with very low power (~20W)	Requires high power (GPUs/TPUs)
Learns from limited data and generalizes well	Needs large data, less generalization
Emotions, consciousness, creativity	No emotions or consciousness (currently)

Inspiration Biologically Inspired AI

Activity 4

How does Face Recognition work?

Expected:

Camera

↓

Image

↓

AI Model

↓

Face Match

Facial recognition is exactly like how your smartphone uses **geometry** and **connect-the-dots** to recognize your face, turning your unique features into a secret, un-hackable password.

Instead of looking at you the way a human friend does, the AI sees your face as a massive math problem.

Here is the step-by-step process of how it works in four simple stages:

1. **Step 1: Face Detection (Finding You)**
2. **Step 2: Facial Analysis (Connect-the-Dots)**
3. **Step 3: Creating your "Faceprint" (The Code)**
4. **Step 4: Making a Match (The Final Verdict)**

Which AI Style Is This?

Facial recognition is a classic example of **Supervised Learning**!

NLP (Natural Language Processing)

Definition

NLP enables computers to understand and generate human language.

Examples:

- ChatGPT
- Google Translate
- Voice Assistants
- Chatbots

NLP Tasks

- Translation
- Text Generation
- Summarization
- Sentiment Analysis
- Question Answering
- Speech Recognition

Live Demo

Open ChatGPT:

Try:

1. "Translate this English sentence into hindi"
2. "Summarize this paragraph"

"Generate Java code"

3. Explain:

This is NLP.

Computer Vision

Definition

Computer Vision enables computers to understand images and videos.

Applications:

Face Unlock

Google Lens

Medical Imaging

OCR

Self-driving Cars

Computer Vision Tasks

- Image Classification
- Object Detection
- Face Recognition
- Image Segmentation
- OCR

Robotics

Definition

Robotics combines AI with mechanical systems.

Components:

Sensors

Camera

Processor

Motors

AI Algorithms

Applications:

- Factory Robots
- Surgical Robots
- Humanoid Robots

TESLA ROBOTS

UNDERSTANDING ROBOTICS IN AI

Tesla robots, also known as Optimus, represent the fusion of Artificial intelligence, Robotics, and Real-World Engineering to build intelligent machines that can perceive, think and act.

AI INTELLIGENCE | PERCEPTION SENSORS | ACTION MECHANISMS | LEARNING & ADAPTATION

Tesla Optimus (Gen 2)

- Height: Height: ~5'8" (172 cm)
- Weight: Weight: ~57 kg
- Payload: Payload: ~20 kg
- Speed: ~5 mph (8 km/h)
- DoF: >40 (Degrees of Freedom)
- AI Powered

WHAT CAN TESLA ROBOTS DO?


Factory Work


Package Handling


Household Tasks


Assistance & Care


Dangerous Tasks


Future Possibilities

HOW TESLA ROBOTS WORK

A. AI BRAIN

1. PERCEPTION

Cameras, sensors and AI vision systems understand the environment.

- 8 Cameras • 3D Vision • LiDAR (optional)

2. AI BRAIN

2. AI BRAIN

Neural networks process data, make decisions and plan actions in real-time.

- FSD Computer • Neural Networks
- Reinforcement Learning

3. ACTION

3. PLANNING & CONTROL

AI plans movements and controls motors to perform tasks smoothly and safely.

- Motion Planning • Path Planning
- Balance & Coordination

4. ACTION

4. ACTION

High precision actuators and hands interact with the real world.

- 28 Actuators • Dexterous Hands
- Force Sensing

KEY COMPONENTS

- **Head Unit** (Cameras, AI Sensors)
- **AI Computer** (FSD Chip)
- **Torso** (Battery, Cooling, Actuators)
- **Arms** (7 DoF Each)
- **Hands** (11 DoF Each Hand)
- **Legs** (12 DoF Each)
- **Feet** (Force Sensors)

AI TECHNOLOGIES POWERING OPTIMUS

Computer Vision → Deep Learning → Reinforcement Learning → SLAM (Mapping) → Natural Language Understanding

HOW TESLA ROBOTS LEARN

Real World Data Collection → Neural Network Training → Simulation & Testing → Deployment & Continous Learning

IMPACT OF TESLA ROBOTS

- ✓ Increases productivity
- ✓ Handles repetitive & dangerous tasks
- ✓ Helps in solving labor shortages
- ✓ Improves quality of life
- ✓ Brings us closer to an AI-powered future

THE FUTURE VISION

Autonomous Factories → Smart Homes → Elderly Care → Space Exploration → Human-Robot Collaboration → A Better & Safer Future

"AI gives robots the mind. Engineering gives them the body. Purpose gives them the future." — Tesla

Generative AI

Definition

Generative AI creates new content.

- Creates:
- Text
- Images
- Videos
- Audio
- Code
- Presentations

Examples:

- ChatGPT
- Gemini
- Claude
- GitHub Copilot

How Generative AI Works

Huge Data

↓

Deep Learning

↓

Large Language Model

↓

User Prompt

↓

Generated Output

Example:

Activity:- codex

Prompt:

"Create Java calculator program"

Output:

AI Generated Code

Traditional AI vs Generative AI

Traditional AI	**Generative AI**
Classifies	Creates
Predicts	Generates
Detects	Builds
Analyses	Designs

Example:
Spam Filter → Traditional AI
ChatGPT → Generative AI

Real World Example: Self Driving Car (Complete Part)

Camera

↓

Computer Vision

↓

Detect Objects

↓

Machine Learning

↓

Predict Movement

↓

AI Decision

↓

Robotics

↓

Brake / Accelerate

AI Technology Mapping Challenge

- Divide students into groups.

Give applications:

YouTube Recommendation

Face Unlock

ChatGPT

Self Driving Car

Example:-

Amazon Recommendation

Students identify:- ?

AI technology involved.:- ?

Homework

AI Exploration Task

Find:

* One AI tool other than ChatGPT.
* What problem it solves.
* Which AI technology it uses.

Example:

Google Lens

Technology:- Computer Vision

Thank You

Questions and discussions